Stereo and Foveated Rendering

@Luiz Velho with ChatGPT

> `Luiz: I am questioning whether biological inspired stereo 3D reconstruction makes sense - and if it does whether the shortcomings are due to the choice of a wrong approach based on traditional stereo`

> `What is your opinion?`

I think your retrospective is basically right, with one important qualification:

the project exposed a mismatch between foveated rendering and the particular way depth was being reconstructed.

The RGB result and the depth result are fundamentally different problems.

1. Why RGB can work while depth fails

For RGB, stochastic ray tracing is almost ideally matched to foveation. A coarse peripheral sample represents the radiance integrated over a larger angular footprint. If neighboring colors are averaged or reconstructed, the result can still be perceptually convincing.

Depth is different.

Suppose a coarse retinal cell straddles the edge of a foreground object:

```
      cell footprint
   +-----------------+
   | foreground |    |
   |------------|    |
   | background      |
   +-----------------+
```

There may be **two physically valid depths inside that one cell**.

For RGB, averaging those samples produces antialiasing.

For depth, $z_{\rm foreground}=2\,{\rm m}, \quad z_{\rm background}=10\,{\rm m} $ does **not** imply that a value such as $z=6\,{\rm m}$ has any physical meaning.

This is, I think, one of the fundamental issues uncovered by the experiments.

A variable-resolution retinal element should perhaps not contain $\boxed{z}$, at all.

It should contain something more like $\boxed{p(z)}$ or even several competing surface hypotheses.

That is a much more important distinction than changing one block matcher for another.

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2. Phase D is strong evidence for this interpretation

The Phase D results you just obtained point in exactly this direction.

The original interpretation of the large depth errors was roughly:

> the stereo matcher is choosing many incorrect disparities.

But the diagnosis showed that most cells failing the 25% depth criterion had actually selected the **correct matching peak**. Their depth was simply too coarsely determined by the angular sampling resolution.

That is exactly what we should expect geometrically.

For conventional rectified stereo,

$Z = \frac{fB}{d}$.

To first order,

$\frac{\Delta Z}{Z} \approx \frac{\Delta d}{d}$.

In angular terms, if binocular parallax is p,

$p \approx \frac{B}{Z}$,

so approximately

$\frac{\Delta Z}{Z} \approx \frac{\Delta p}{p}$.

Now suppose the peripheral retinal cell has angular width $\Delta\theta$. You cannot determine parallax much more accurately than the information contained inside that angular sampling interval.

So there is roughly an intrinsic floor of the form

$\frac{\Delta Z}{Z} \gtrsim \frac{\Delta \theta}{p}$.

As eccentricity increases,

$\Delta\theta \uparrow$,

and for distant objects

$p \downarrow$.

Those two effects work in exactly the wrong direction for precise metric depth.

So the Phase D discovery that a large part of the apparent gross error was really **resolution error rather than incorrect correspondence** is not an implementation accident. It is telling you something fundamental about the formulation.

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3. Does biologically inspired stereo 3D reconstruction still make sense?

Yes—but I would change what "3D reconstruction" means.

I am now much less convinced that the biologically inspired objective should be:

> produce a dense metrically accurate depth image over the entire visual field.

There is little reason to believe that this is what biological vision does.

Your physiology report already points toward a different interpretation. Foveated vision is described as an active system in which peripheral evidence selects targets, saccades reposition the high-resolution sensor, and successive fixations support task-dependent understanding. Stereoacuity is also best near fixation and degrades with eccentricity.

And your original project note makes the deeper point that variable-resolution sampling should be treated as a **controlled sampling problem**, with the scientific question becoming whether one can obtain the same useful posterior with less computation.

That suggests that the biologically meaningful output is not necessarily

$$D(x, y)$$

for every image pixel.

It may instead be something like

$$\boxed{\text{a progressively refined, uncertainty-aware 3D model of the scene}}$$

constructed by active observation.

That is a substantially different objective.

4. I think the traditional stereo formulation is probably the wrong *core* approach

I would not say traditional stereo was a bad choice. It was an excellent baseline because it made the experiments understandable.

But it is poorly matched to where the novelty of this project actually lies.

Traditional stereo usually assumes, explicitly or implicitly:

- a regular pixel lattice;
- fairly uniform angular resolution;
- translation-invariant matching windows;
- one disparity per image location;
- predominantly local spatial smoothness;
- two images acquired at essentially the same resolution.

Your artificial retina violates several of these assumptions.

The sampling lattice is spatially varying. A retinal element has a finite angular footprint whose size changes with eccentricity. At object boundaries that footprint may contain multiple surfaces. And repeated fixations observe the same part of the world under different retinal sampling geometries.

Your earlier project note actually anticipated this: a matcher operating directly on a non-uniform lattice was identified as a **research problem rather than an engineering modification of ordinary stereo**.

Phase D strengthens that conclusion.

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5. The consensus experiment may be pointing toward the right representation

One of the most interesting Phase D observations was:

$$P(\text{second bad} \mid \text{first bad}) \gg P(\text{second bad} \mid \text{first good}).$$

In other words, errors are systematic rather than independent.

That is why simple Gaussian inverse-variance averaging eventually hurts.

And when two fine observations disagree, one of them is overwhelmingly likely to be wrong.

Your `ConsensusBelief` is already taking a small step away from the classical formulation.

Instead of

$$\rho_1, \rho_2, \rho_3 \rightarrow \text{weighted mean}$$

it effectively asks

$$\{\rho_1, \rho_2, \rho_3\} \rightarrow \text{which hypotheses agree?}$$

I think the logical next step is to stop forcing the state into a single Gaussian at all.

For example, a retinal cell could maintain

$$p(\rho) = \sum_{k=1}^{K} w_k \, \mathcal{N}(\rho; \mu_k, \sigma_k^2).$$

Near a clean planar surface perhaps

$$K = 1.$$

At a depth boundary,

$$K = 2$$

might represent foreground and background surfaces.

If subsequent fixation resolves the ambiguity, one component gains probability.

That would turn the current idea of an **undecided cell** into a first-class perceptual state rather than a failure condition.

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6. There are really two projects hiding inside the current one

I would separate them conceptually.

Goal	Appropriate depth mechanism
Foveated photorealistic rendering	Renderer-derived geometric depth
Biologically inspired active perception	Inferred uncertain/multihypothesis depth

For the first problem, stereo reconstruction is unnecessary.

Blender already knows exactly where the primary ray intersects the scene.

So every primary ray can produce simultaneously

$$(\mathbf{c},\ z,\ \mathbf{n},\ \mathbf{x},\ \text{object ID}, \ldots)$$

where **c** is radiance and z is geometric depth.

The real research problem then becomes:

> How do we reconstruct a useful full-resolution depth representation from a nonuniform collection of exact geometric samples?

That itself is interesting, particularly around discontinuities.

But you should not have to solve correspondence to make a foveated renderer produce depth.

This is consistent with the rendering architecture we discussed earlier: spatial sampling density and Monte Carlo sampling effort can be controlled independently inside the stochastic renderer.

For the **second** project, however, direct access to renderer depth is forbidden except as training/evaluation ground truth.

That becomes the active-vision research program.

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7. I would also change the biological question

Instead of asking:

> Can a foveated binocular camera reconstruct an accurate depth image?

I would ask:

> **Can a foveated active binocular observer acquire sufficient 3D structure for a task using substantially less visual information than a uniform-resolution stereo system?**

That change is subtle but important.

The benchmark would no longer simply be

$$\mathrm{RMSE}(D, \hat{D}).$$

You could measure things such as:

- accuracy of foveated surface geometry;
- object boundary localization;
- object-scale 3D reconstruction;
- completeness as a function of gaze budget;
- uncertainty calibration;
- number of fixations required to resolve ambiguous regions;
- reconstruction quality versus rays/computation;
- semantic or task performance from the reconstructed model.

That is much closer to the biological motivation.

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8. And I would not throw stereo away

Stereo remains extremely useful.

But I would demote it from

> **the mechanism that reconstructs the scene**

to

> **one measurement process contributing evidence about 3D structure.**

The system could ultimately use several cues:

$$
\begin{aligned}
&\text{binocular disparity}\\
+&\text{vergence}\\
+&\text{motion parallax}\\
+&\text{focus/accommodation}\\
+&\text{occlusion}\\
+&\text{surface continuity}\\
+&\text{semantic priors.}
\end{aligned}
$$

That feels much more biologically plausible.

And importantly, the active system controls some of those measurements.

If the depth posterior at a location is bimodal,

$$p(z) = w_1 p_1(z) + w_2 p_2(z),$$

the observer can deliberately make another fixation or change vergence to acquire a measurement that discriminates between the two hypotheses.

Then gaze has a genuinely computational role:

$$\boxed{\text{choose the next observation that reduces 3D uncertainty}}$$

rather than simply

$$\text{look somewhere that has not been seen yet.}$$

That would be a substantial conceptual advance over Phase C.

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9. If stereo remains central, I see three possible directions

I would consider them in this order.

First, local foveal stereo. Use conventional or learned high-quality stereo only in a relatively small region around fixation. Project that region onto a locally uniform tangent image, perform stereo there, and insert those depth estimates into the persistent 3D model. This is practical and provides a strong baseline.

Second, direct probabilistic 3D fusion. Instead of constructing a global disparity image, convert each reliable binocular observation into a 3D hypothesis and fuse successive observations in world coordinates. This is the direction I find most conceptually attractive for the project.

Third, a genuinely non-uniform learned retinal matcher. Represent retinal samples by their ray directions, angular footprints, image features, and eye identity, and learn correspondence without first resampling onto a uniform grid. A graph/set/attention architecture could produce a likelihood over inverse depth rather than one disparity. Blender provides essentially unlimited exact ground truth for training it.

That third direction is the most ambitious research problem, but it is also the one most faithful to the original idea.

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10. The biggest conceptual lesson from the current project

I think Phase D revealed something more important than “the block matcher isn't good enough.”

It revealed that:

$$\boxed{\text{foveated depth is not simply low-resolution conventional depth}}$$

The same operation that works nicely for RGB—

$$\text{reduce spatial sampling and reconstruct}$$

—does not transfer directly to scene geometry.

Geometry has discontinuities, occlusions, layered surfaces, and strong nonlinear consequences from angular uncertainty.

A retinal cell therefore probably should represent something closer to:

$$\{\text{direction}, \text{angular footprint}, \text{appearance}, p(\rho), \text{surface hypotheses}, \text{confidence}\}$$

than

$$\{\text{RGB}, \text{depth}\}.$$

That, to me, is where the research opportunity now lies.

My overall assessment

I would **not abandon biologically inspired 3D vision**.

I would abandon the requirement that it behave like a conventional dense stereo algorithm operating at varying resolution.

The strongest ideas from the project remain very good:

$$\text{foveation} + \text{variable resolution} + \text{binocular sensing} + \text{active gaze} + \text{vergence} \\ + \text{sequential evidence}.$$

What I would replace is the middle:

$$\cancel{\text{block stereo} \rightarrow \text{single disparity} \rightarrow \text{Gaussian averaging}}$$

with something more like

$$\boxed{\text{stereo evidence} \rightarrow \text{multihypothesis 3D belief} \rightarrow \text{active re-observation} \rightarrow \text{progressive scene model.}}$$

In fact, the fact that the current approach has now exposed this limitation may be the most scientifically useful result of Phases C–D. It points toward a project that is **more distinctly about active biological vision**, rather than a foveated implementation of classical stereo.

> `Luiz : Can you review the papers below and assess their relationship with the ideas we are discussing?`

https://pmc.ncbi.nlm.nih.gov/articles/PMC10526497/
https://www.pnas.org/doi/full/10.1073/pnas.1908100117
https://jov.arvojournals.org/article.aspx?articleid=2121400
https://www.nature.com/articles/srep44800

Yes. After reading all four, I would modify my previous assessment in an important way:

These papers make me more—not less—convinced that biologically inspired binocular 3D vision is a worthwhile direction. But they also make me less convinced that the right computational formulation is "traditional stereo matching applied to foveated images, followed by a dense depth map."

Taken together, the papers suggest a rather different architecture: **stereo as probabilistic evidence, adapted to natural disparity statistics and retinal eccentricity, embedded inside an active perception–action loop.** None of the four papers, however, actually demonstrates dense, metrically accurate depth reconstruction of a complex scene from a foveated binocular sensor. That distinction is crucial.

How the four papers fit together

Paper	Main idea	Relation to your project
Priorelli et al., 2023	Depth through **active inference** and fixation with nonuniform foveal resolution	Very close to your high-level concept
Eckmann et al., 2020	**Sensory representation and eye movements co-adapt** through active efficient coding	Strong support for jointly learning representation + vergence/action
Burge & Geisler, 2014	Natural stereo should be decoded as a **probabilistic population estimate of disparity**	Most relevant candidate replacement for your NCC/block matcher
Gibaldi et al., 2017	Actual fixation profoundly changes the **statistics of disparity with eccentricity**	Strong argument against a spatially uniform disparity matcher/search rule

The particularly interesting point is that these four papers occupy different layers of what could become a new architecture for your project.

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1. Priorelli, Pezzulo & Stoianov — *Active Vision in Binocular Depth Estimation: A Top-Down Perspective*

This is probably the paper that is conceptually closest to where our discussion has just arrived.

The authors explicitly reject depth estimation as merely a bottom-up feature-detection process. Instead, they formulate it as **active inference**: a hierarchical generative model predicts the retinal projections of an object, and depth is inferred by reducing prediction errors. Eye orientation and depth estimation are solved together through repeated perception–action cycles. (PubMed Central (PMC))

The most striking result for your project is their comparison between **uniform and nonuniform retinal resolution**.

With a nonuniform fovea, fixation matters: as the two eyes converge on the target and place it near their high-resolution centers, depth estimation improves dramatically. Their active agent, which simultaneously estimates depth and moves its eyes, almost reaches the performance of an agent whose eyes are already correctly fixated. But when retinal resolution is made uniform, active fixation provides no advantage and can even be detrimental. (PubMed Central (PMC))

That is remarkably close to your original premise:

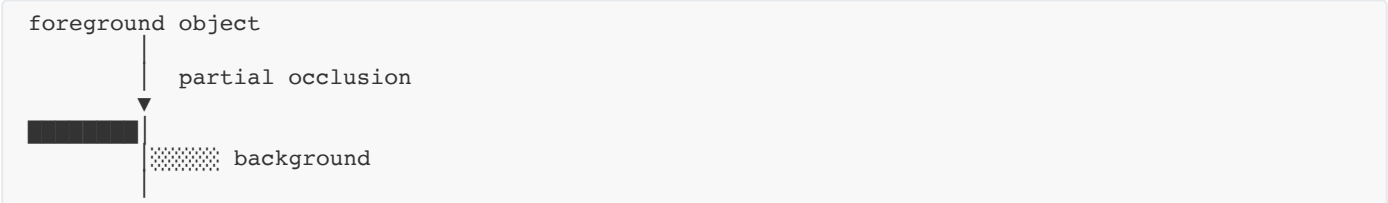
$$\text{foveation} + \text{vergence} + \text{active gaze} \longrightarrow \text{better depth information}.$$

It also agrees with the biological argument in your eye report that foveation only really makes sense as part of an active sensorimotor system.

But there is a very important limitation

Priorelli et al. do **not** solve the stereo correspondence problem faced by your system.

Their task is essentially inference of the position/depth of a target through a generative projection model. They don't have to deal with something like:



where one coarse retinal receptive field contains multiple surfaces and the left and right observations contain different visible subsets of those surfaces.

So I would read this paper as very strong support for your **system-level concept**, but not as evidence that its particular computational mechanism will solve your current depth problem.

In fact, it indirectly reinforces our previous conclusion:

> The active binocular system should infer a 3D belief; it need not first construct a conventional dense disparity image.

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2. Eckmann et al. — *Active efficient coding explains the development of binocular vision and its failure in amblyopia*

This paper is less directly about 3D reconstruction, but in some ways it is even more important architecturally.

Its central principle is **Active Efficient Coding (AEC)**:

> sensory coding and visual behavior should be optimized together because behavior determines the statistics of the sensory signals that the system receives.

Their system contains retinal preprocessing, binocular cortical encoding, and reinforcement-learning controllers for **vergence and accommodation**. The cortical representation and motor policies develop simultaneously rather than assuming that one is already calibrated before learning the other. (PubMed Central (PMC))

Even more relevant to your work, they explicitly use distinct high-resolution foveal and lower-resolution peripheral encoding. The foveal pathway supports fine binocular behavior while peripheral information remains useful for coarse control. (PubMed Central (PMC))

Their key feedback loop can be summarized as

$$\text{current representation} \to \text{vergence action} \to \text{new binocular statistics} \to \text{better representation} \to \cdots$$

This is quite different from the architecture of the current `fov-3d-vision` implementation, where broadly speaking:

$$\boxed{\text{matcher is fixed} \to \text{depth measurements} \to \text{belief} \to \text{gaze policy}}$$

AEC suggests that the first and last stages should perhaps **co-adapt**.

What the paper does not demonstrate

It does not reconstruct complex 3D scenes.

Their visual environment is deliberately simplified: natural image texture is moved in disparity and blurred to simulate vergence and accommodation, and the objective is efficient representation plus accurate motor control. (PubMed Central (PMC))

So it would be incorrect to conclude:

> AEC provides a replacement stereo reconstruction algorithm.

It doesn't.

But it provides a powerful design principle:

$$\boxed{\text{Don't optimize the depth estimator independently of the actions generating its input.}}$$

That principle may be very important for the next version of your project.

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3. Burge & Geisler — *Optimal disparity estimation in natural stereo images*

This is the paper I now find **most relevant to the immediate technical problem in your implementation**.

Burge and Geisler ask a normative question:

> Given natural binocular image statistics and the properties of the visual sensor, what computations should optimally estimate disparity?

They do not simply perform correlation and choose its maximum.

Their pipeline is approximately:

$$\text{photoreceptor signals} \rightarrow \text{learned binocular filters} \rightarrow \text{nonlinear disparity-selective population} \rightarrow p(d \mid \text{responses}) \rightarrow \text{disparity estimate}.$$

The binocular filters are learned specifically to maximize disparity-estimation accuracy on natural images. The resulting filters resemble properties of binocular V1 neurons. Crucially, the intermediate representation is a **population encoding a probability distribution over disparity**, and the MAP disparity is just one possible readout. (PubMed Central (PMC))

For their estimator,

$$p(d \mid R) \propto p(R \mid d)\, p(d),$$

where R denotes the binocular filter responses.

They explicitly note that other readout rules besides MAP are appropriate for other cost functions and that nonuniform disparity priors can be incorporated. (PubMed Central (PMC))

This connects directly to what Phase D discovered

The current pipeline roughly does:

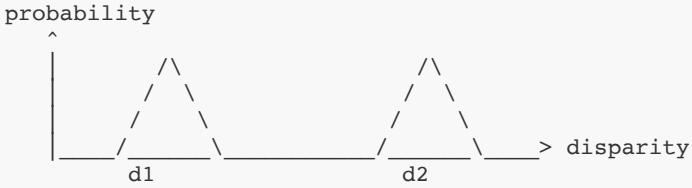
$$\text{matching response} \rightarrow d \rightarrow \sigma_d^2$$

and subsequently treats that as something close to a Gaussian measurement.

Phase D is telling you that this loses too much information.

Imagine a matching response like this:

$p(d)$



Choosing

$$d = d_1$$

and attaching a variance cannot express the existence of the competing d_2 hypothesis.

Burge and Geisler give a very principled reason to retain the **entire disparity evidence distribution**.

This fits extremely well with what your `ConsensusBelief` began to uncover.

Instead of:

$$d_1, d_2, d_3 \rightarrow \text{weighted average},$$

you could eventually have:

$$p_1(d), p_2(d), p_3(d) \rightarrow p(d \mid \text{all observations}).$$

That is a much more natural framework for repeated fixation.

Their result is also bad news for simple block correlation

They compare their task-optimized encoding with local cross-correlation and report advantages particularly as disparities become larger or vary across the patch. They explicitly state that cross-correlation is optimal only under restricted circumstances where disparity is small and nearly constant across the patch. (PubMed Central (PMC))

That is very relevant to your system because your peripheral cells do exactly the opposite:

$$\text{larger footprint} \Rightarrow \text{greater chance of disparity variation within the footprint}.$$

But again there is a caveat

Their main experiment uses approximately $1^\circ \times 1^\circ$ patches derived from **locally planar surfaces**, near the fovea, with disparities inside the fusional range. (PubMed Central (PMC))

Therefore, they largely sidestep the hardest case in your renderer:

$$\boxed{\text{one retinal element containing multiple depth surfaces}}$$

plus occlusions.

So I see this paper not as the complete solution, but as an excellent basis for replacing your current **NCC peak → scalar disparity** model.

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4. Gibaldi, Canessa & Sabatini — *The Active Side of Stereopsis*

Of the four, this paper may have the strongest implications for the **geometry of your foveated matcher**.

They combine realistic 3D scenes with actual binocular eye-movement recordings and ask:

> What disparity statistics does an observer actually experience when looking naturally?

The answer differs substantially from what is obtained using random fixation.

Human observers preferentially fixate behaviorally relevant nearby objects. As a consequence, the disparity distribution around the current fixation becomes highly structured: the central region tends toward small disparity while the surrounding scene often lies farther away. (PubMed Central (PMC))

They find that actual fixation statistics predict known properties of binocular vision better than random viewing. In particular, the disparity range and effective receptive-field scale increase with retinal eccentricity, while the central visual field supports a much narrower disparity range. (Nature)

This has a very direct implication for your matcher.

The search space should probably **not** be

$d \in [-D, D]$

everywhere.

It should look more like

$p(d \mid e, g, \text{context}),$

where

- e = eccentricity,
- g = gaze/fixation state,
- context includes expected foreground/background geometry.

The paper even discusses how natural eye posture and binocular Listing's law help align the retinal meridians and thereby **reduce the correspondence search region**. (PubMed Central (PMC))

This makes your earlier project observation particularly prescient: a matcher operating on the nonuniform retinal lattice is probably a genuine research problem rather than an engineering modification to standard stereo.

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5. The four papers form a surprisingly coherent architecture

I think they can almost be stacked on top of one another.

Gibaldi et al.: what should the prior be?

Natural active fixation determines:

$$\boxed{p(d \mid e, g, \text{scene context})}$$

Disparity statistics depend on where the observer is looking and on retinal eccentricity.

Burge & Geisler: what should the sensory likelihood be?

Binocular receptive fields encode evidence:

$$\boxed{p(R_L, R_R \mid d)}$$

and produce a population posterior rather than simply a correlation maximum.

Eckmann et al.: how should the sensor and action policy develop?

Representation and action should be optimized together:

$$\boxed{\text{encoding} \leftrightarrow \text{vergence/accommodation/action}}$$

rather than designed independently.

Priorelli et al.: what should the overall computational loop be?

Depth emerges through:

$$\boxed{\text{prediction} \rightarrow \text{observation} \rightarrow \text{prediction error} \rightarrow \text{eye movement} \rightarrow \text{better observation.}}$$

Put together, this is almost exactly the direction we were reaching independently in our previous discussion.

6. It changes how I see the shortcomings of your current system

Previously I characterized the problem as perhaps being that a traditional stereo algorithm was poorly suited to a nonuniform lattice.

After these papers, I would be more specific.

I think **three assumptions of the current formulation are questionable**.

A. One disparity per retinal cell

This is probably the most serious one.

At a depth boundary a coarse cell can legitimately contain:

$$d_1, \quad d_2,$$

not an intermediate

$$\bar{d}.$$

The state should therefore allow something like

$$p(d)$$

or

$$p(d) = \sum_k w_k p_k(d),$$

possibly with an explicit occlusion/no-correspondence hypothesis.

B. Stationary matching statistics

A traditional matcher assumes roughly the same matching process everywhere.

The biological literature instead strongly suggests:

$$p(d), \quad \text{receptive-field scale}, \quad \text{precision}$$

all vary with eccentricity and fixation. Gibaldi et al. make this especially clear. (PubMed Central (PMC))

Therefore the matcher itself should be **retinotopically conditioned**.

C. Observations are independent measurements of the same scalar

Your Phase D experiments already challenged this.

A repeated bad measurement tends to remain bad. That is exactly what one expects if the ambiguity originates from structural factors such as an occlusion, mixed receptive field, or repeated incorrect correspondence.

The proper state is then not

$$\rho \sim \mathcal{N}(\mu, \sigma^2)$$

with every fixation adding another Gaussian observation.

It is closer to:

$$B_t = \{H_1, H_2, \dots, H_K; w_1, \dots, w_K\},$$

where H_k are competing scene hypotheses.

A new fixation changes their probabilities.

That interpretation is very compatible with all four papers, although none implements precisely this model.

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7. This also clarifies what “depth” should mean in your project

I would make a stronger distinction than before.

For the **foveated Blender renderer**, if the objective is to output an accurate depth pass, I see little scientific reason to infer it from stereo. Cycles already knows the primary-ray intersection:

$$\mathbf{x} = \mathbf{o} + t\mathbf{d}.$$

That gives exact geometric depth.

The interesting rendering problem is then:

> How should exact but nonuniformly sampled geometry be reconstructed into an appropriate variable-resolution depth representation?

That is one research problem.

The biologically inspired project is a different problem:

> Given only the sensory evidence available to the two artificial eyes, how should an active observer construct an increasingly useful 3D belief?

That is where these four papers become highly relevant.

I would keep the renderer's geometric depth as **ground truth**, not as sensory input.

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8. I would therefore reconsider the target representation

The current conceptual state might be something like

$$B(x, y) = (\rho, \sigma_\rho).$$

I would instead consider something closer to

$$B(x, y) = [p(\rho), P_{\text{occ}}, P_{\text{unresolved}}, \text{support}] .$$

And ultimately I would not even make the retinal lattice the permanent world representation.

Each fixation gives observations in retinal coordinates:

$$p(\rho \mid R_L, R_R, g_t).$$

Those observations should be transformed into a persistent **world-centered 3D belief**:

$$\mathcal{B}_t(\mathbf{X}).$$

Then:

$$\mathcal{B}_{t+1} = \text{Update}\left(\mathcal{B}_t, p(\rho \mid R_L, R_R, g_t)\right) .$$

That means the retina is the **sensor**, rather than the map.

This distinction may solve several conceptual problems in the current approach.

9. And then the next fixation has a mathematically clear purpose

Instead of coverage-first:

$g_{t+1} = \arg\max_g \text{unseen area},$

the observer could ask:

$$\boxed{g_{t+1} = \arg\max_g \frac{\mathbb{E}[H(\mathcal{B}_t) - H(\mathcal{B}_{t+1})]}{C(g)}}$$

where

- H is uncertainty/entropy,
- $C(g)$ is the computational or motor cost of the observation.

In words:

> Look where another fixation is expected to reduce uncertainty the most per unit cost.

That finally gives the fovea a fundamental computational reason to exist.

And your Phase D `undecided` cells are almost a primitive version of exactly this idea.

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10. I would not jump directly to a large neural network

Burge & Geisler suggest an appealing intermediate research step.

Blender can generate enormous amounts of controlled ground truth:

$$(R_L, R_R, d_{\mathrm{GT}}, e, \mathrm{occlusion}, \mathrm{surface\ IDs}, \ldots).$$

You could train a relatively compact **task-optimal binocular encoder** modeled after their approach—or a modern lightweight equivalent—to output:

$$p(d \mid R_L, R_R, e).$$

The experiment I would try first is:

$$\boxed{\mathrm{NCC\ peak} \quad \mathrm{vs.} \quad \mathrm{learned\ disparity\ posterior}}$$

on exactly the Phase C/D observation records.

I would stratify the evaluation into:

- single-surface cells;
- depth-boundary cells;
- half-occlusion cells;
- foveal vs parafoveal vs peripheral;
- first look vs repeat look.

The most interesting question would not initially be whether the new method lowers global RMSE.

It would be:

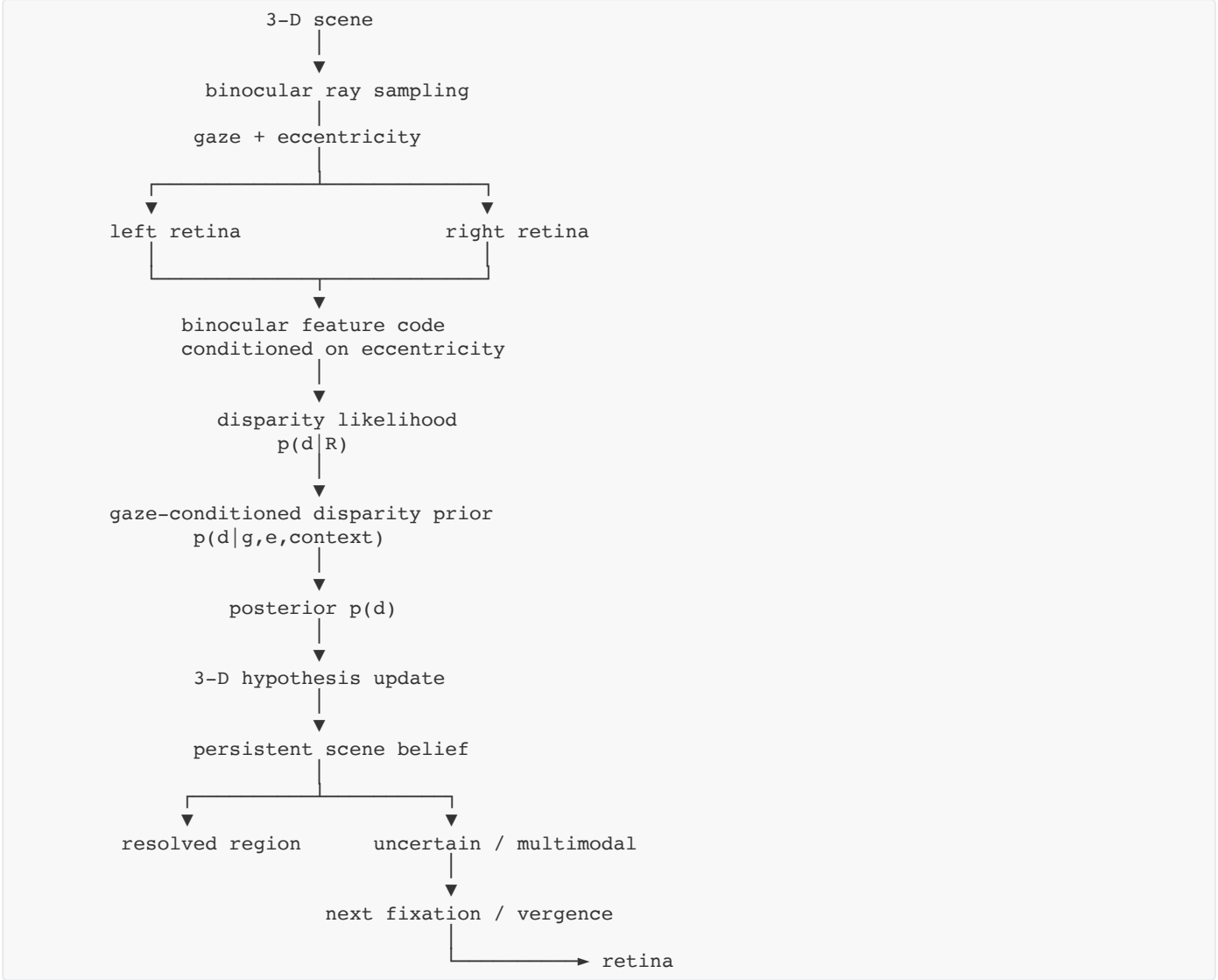
> **Does its posterior know when the correspondence is ambiguous?**

That is much more important for active perception.

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11. A possible revised architecture

After these papers, this is the architecture I would now investigate:



Notice what has disappeared:

$$\text{uniform block match} \Rightarrow \cancel{\text{one disparity}} \rightarrow \text{depth image}$$

Stereo is still there—but its role has changed.

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12. The most important conclusion from the four papers

I don't think the literature supports the conclusion:

> "Biological stereo is the wrong direction."

I think it supports almost the opposite:

> **Biological stereo may be a very promising direction, but biological stereo is not equivalent to classical dense stereo performed on two nonuniform images.**

The biological picture emerging from these papers is:

$$\boxed{\begin{aligned} &\text{nonuniform sensing} \\ +\,&\text{population disparity coding} \\ +\,&\text{natural-scene priors} \\ +\,&\text{vergence} \\ +\,&\text{saccades} \\ +\,&\text{adaptive receptive fields} \\ +\,&\text{sequential inference} \end{aligned}}$$

rather than

$$\boxed{\text{left image} + \text{right image} \rightarrow \text{correspondence} \rightarrow \text{dense disparity map.}}$$

And there is a particularly useful division of labor among these papers:

Burge & Geisler tell us how binocular evidence might be encoded. Gibaldi et al. tell us how gaze changes its prior statistics. Eckmann et al. tell us that encoding and action should co-adapt. Priorelli et al. tell us how the whole thing can be regarded as an active inference loop.

None solves your full problem. But together they suggest a much more coherent path forward than trying to improve the present block matcher incrementally.

If I were deciding the next research phase now, I would make the central question:

> **Can an eccentricity-conditioned probabilistic binocular representation, coupled to active fixation, produce a better calibrated and more useful 3D scene belief per ray than conventional foveated stereo?**

That question preserves the most original parts of your project—variable resolution, foveation, binocular vision, vergence and active gaze—while directly addressing what Phases C and D have taught us about the weaknesses of the traditional stereo formulation.
